NexLearn

Learning Management System

Use Case Model & Specifications

UE23CS352B – Object Oriented Analysis & Design

Domain: Education

Technology: Spring Boot 3.4 + Next.js 15 + MongoDB

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1. Project Overview

NexLearn is a full-stack Learning Management System (LMS) built for the Education domain. It enables instructors to design and publish courses, and students to browse, enroll, and track their learning progress. The system is implemented using Spring Boot 3.4 (MVC backend with JWT security), Next.js 15 (React frontend), and MongoDB (NoSQL persistence).

1.1 Actors

Actor	Description
Student	Browses courses, enrolls in courses, accesses content, and tracks learning progress through the Dashboard.
Instructor	Creates and manages courses, uploads educational content, and monitors enrolled student progress.
System (Spring Boot)	Validates requests, enforces JWT authentication, and persists data in MongoDB via Spring Data.

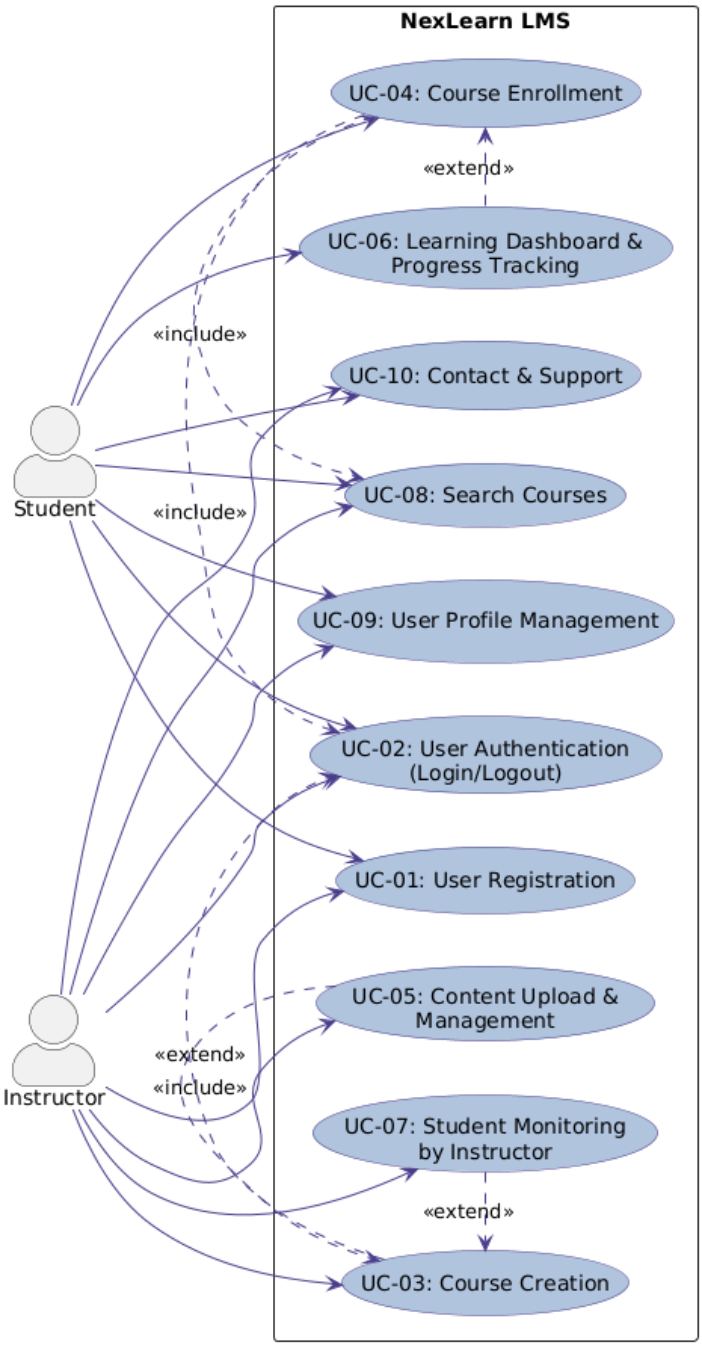
1.2 Use Case Summary

UC ID	Use Case Name	Primary Actor	Priority	Type
UC-01	User Registration	Student, Instructor	High	Major
UC-02	User Authentication (Login/Logout)	Student, Instructor	High	Major
UC-03	Course Creation	Instructor	High	Major
UC-04	Course Enrollment	Student	High	Major
UC-05	Content Upload & Management	Instructor	High	Major
UC-06	Learning Dashboard & Progress Tracking	Student	Medium	Minor
UC-07	Student Monitoring by Instructor	Instructor	Medium	Minor
UC-08	Search Courses	Student, Instructor	Medium	Minor
UC-09	User Profile Management	Student, Instructor	Low	Minor
UC-10	Contact & Support	Student, Instructor	Low	Minor

2. Use Case Diagram

The Use Case Diagram for NexLearn

Use Case Diagram



3. Use Case Specifications

The following section provides detailed use case specifications for all 10 use cases (4 major, 4 minor is the minimum; NexLearn covers 5 major and 5 minor for completeness).

UC-01 – User Registration

Use Case ID	UC-01
Use Case Name	User Registration
Primary Actor(s)	Student, Instructor
Description	A new user creates an account in NexLearn by providing their details and selecting a role.
Priority	High
Type	Major
Pre-conditions	User is not already registered in the system.
Post-conditions	A new user account is created and stored in MongoDB. The user can now log in.
Main Flow	1. User navigates to the Registration page. 2. User enters full name, email, password, and selects role (Student/Instructor). 3. System validates input fields for completeness and email format. 4. System checks that the email is not already registered. 5. System hashes the password and creates a new user record in MongoDB. 6. System displays a success message and redirects to Login page.
Alternative / Exception Flow	3a. If any field is missing or malformed, system displays validation error. 4a. If email already exists, system displays 'Email already in use' error.

UC-02 – User Authentication (Login/Logout)

Use Case ID	UC-02
Use Case Name	User Authentication (Login/Logout)
Primary Actor(s)	Student, Instructor
Description	A registered user logs into the system with their credentials and receives a JWT token for session management.

Priority	High
Type	Major
Pre-conditions	User has an existing account.
Post-conditions	User is authenticated; a JWT token is issued and stored client-side. Dashboard is accessible.
Main Flow	1. User navigates to the Login page. 2. User enters registered email and password. 3. System validates credentials against the MongoDB user record. 4. System generates a signed JWT token and returns it to the client. 5. Client stores the token; user is redirected to role-appropriate Dashboard. 6. User can log out by clicking 'Logout'; token is cleared from client.
Alternative / Exception Flow	3a. If credentials are invalid, system displays 'Invalid email or password'. 4a. If account is inactive/deleted, system denies access.

UC-03 – Course Creation

Use Case ID	UC-03
Use Case Name	Course Creation
Primary Actor(s)	Instructor
Description	An authenticated instructor creates a new course with title, description, category, and initial content.
Priority	High
Type	Major
Pre-conditions	Instructor is logged in with a valid JWT token.
Post-conditions	A new course record is persisted in MongoDB and is visible on the course browsing page.
Main Flow	1. Instructor navigates to 'Create Course' from the Dashboard. 2. Instructor fills in: course title, description, category, and optional thumbnail. 3. Instructor clicks 'Publish'. 4. System validates all required fields. 5. System saves the course record linked to the Instructor's user ID. 6. System redirects Instructor to the Course Management page with a success notification.
Alternative / Exception Flow	4a. If required fields are missing, system highlights the fields and shows error messages. 5a. If a database error occurs, system shows a generic error and does not save.

UC-04 – Course Enrollment

Use Case ID	UC-04
Use Case Name	Course Enrollment
Primary Actor(s)	Student
Description	A student browses available courses and enrolls in a course of their choice.
Priority	High
Type	Major
Pre-conditions	Student is authenticated. The course exists and is published.
Post-conditions	An enrollment record is created; the course appears in the Student's 'My Courses' dashboard.
Main Flow	1. Student navigates to 'Browse Courses'. 2. Student searches/filters courses by name, category, or instructor. 3. Student selects a course to view its details. 4. Student clicks 'Enroll'. 5. System checks the student is not already enrolled. 6. System creates an enrollment record in MongoDB. 7. System redirects Student to the course content page.
Alternative / Exception Flow	5a. If student is already enrolled, system shows 'Already Enrolled' and directs them to the course. 6a. If a server error occurs, system notifies the student and does not create a duplicate.

UC-05 – Content Upload & Management

Use Case ID	UC-05
Use Case Name	Content Upload & Management
Primary Actor(s)	Instructor
Description	An instructor adds, edits, or removes lectures and materials within an existing course.
Priority	High
Type	Major
Pre-conditions	Instructor is authenticated and owns the course.
Post-conditions	Course content is updated in MongoDB and is immediately available to enrolled students.

Main Flow	1. Instructor navigates to 'Manage Course' for a specific course. 2. Instructor clicks 'Add Lecture' and provides a title and content (text/link). 3. System saves the new lecture linked to the course ID. 4. Instructor can edit lecture titles/content or delete individual lectures. 5. System persists all changes to MongoDB.
Alternative / Exception Flow	2a. If content fields are empty, system shows a validation error. 4a. If instructor attempts to delete a course with enrolled students, system shows a warning.

UC-06 – Learning Dashboard & Progress Tracking

Use Case ID	UC-06
Use Case Name	Learning Dashboard & Progress Tracking
Primary Actor(s)	Student
Description	A student views their enrolled courses and tracks overall learning progress.
Priority	Medium
Type	Minor
Pre-conditions	Student is authenticated and enrolled in at least one course.
Post-conditions	Dashboard reflects current enrollment and progress status.
Main Flow	1. Student logs in and navigates to the Dashboard. 2. System retrieves all enrollments associated with the student. 3. System displays course cards with title, instructor, and progress indicator. 4. Student clicks a course to access its content. 5. System marks individual lectures as viewed upon access.
Alternative / Exception Flow	2a. If student has no enrollments, system shows 'Browse Courses' call-to-action. 5a. If a network error occurs, progress update is retried on next access.

UC-07 – Student Monitoring by Instructor

Use Case ID	UC-07
Use Case Name	Student Monitoring by Instructor
Primary Actor(s)	Instructor

Description	An instructor views the list of students enrolled in their courses and monitors their progress.
Priority	Medium
Type	Minor
Pre-conditions	Instructor is authenticated and has at least one published course.
Post-conditions	Instructor has visibility into student enrollment and activity data.
Main Flow	1. Instructor navigates to 'Manage Course' and selects 'Students'. 2. System retrieves all enrollment records for that course. 3. System displays a list of enrolled students with name and progress. 4. Instructor can view individual student progress details.
Alternative / Exception Flow	2a. If no students are enrolled, system displays 'No students enrolled yet'.

UC-08 – Search Courses

Use Case ID	UC-08
Use Case Name	Search Courses
Primary Actor(s)	Student, Instructor
Description	Any authenticated user searches for courses using keywords, category, or instructor name.
Priority	Medium
Type	Minor
Pre-conditions	User is on the Browse Courses page.
Post-conditions	A filtered list of matching courses is displayed.
Main Flow	1. User enters a search term in the search bar or selects a category filter. 2. System queries MongoDB for courses matching the criteria. 3. System renders the matching course cards in real time. 4. User selects a course to view details.
Alternative / Exception Flow	2a. If no matches found, system displays 'No courses match your search'. 3a. If search term is empty, all published courses are displayed.

UC-09 – User Profile Management

Use Case ID	UC-09
Use Case Name	User Profile Management
Primary Actor(s)	Student, Instructor
Description	Users can view and update their personal profile information.
Priority	Low
Type	Minor
Pre-conditions	User is authenticated.
Post-conditions	Updated profile is saved to MongoDB.
Main Flow	1. User navigates to 'Profile' from the navigation bar. 2. User views current name, email, and role. 3. User edits permitted fields (e.g., name, password). 4. User clicks 'Save Changes'. 5. System validates and persists the changes.
Alternative / Exception Flow	3a. If new password doesn't meet policy, system shows error. 5a. If email is changed to one already in use, system rejects the update.

UC-10 – Contact & Support

Use Case ID	UC-10
Use Case Name	Contact & Support
Primary Actor(s)	Student, Instructor
Description	Users submit a support query or feedback via a contact form.
Priority	Low
Type	Minor
Pre-conditions	User is on the Contact/Support page (accessible without authentication).
Post-conditions	Support message is recorded or forwarded for follow-up.
Main Flow	1. User navigates to the 'Contact' page. 2. User fills in name, email, subject, and message. 3. User submits the form. 4. System validates input and acknowledges submission with a success message.
Alternative / Exception Flow	2a. If required fields are missing, system shows validation errors.